
KAN-Parameterized Energy-Based Models: Landscape Geometry and Test-Time Compute Scaling

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Abstract

The dominant deep-learning paradigm computes a fixed-cost function, capping accuracy at model capacity rather than available test-time budget. Energy-based models (EBMs) offer an alternative—iterative gradient descent on a learned energy—but standard global-flatten MLP-EBMs create shallow basins that plateau at $K=1$. We study **KAN-EBM**, which pairs a convolutional filter bank with a Kolmogorov-Arnold Network as a per-pixel energy density. Our main empirical contribution is a *predictive scaling law* for optimal inference depth: $K^*(\sigma) \approx C\sigma^\alpha$ with $\alpha \approx 1.3$ – 1.5 ($R^2=0.97$ – 0.98), reproducible across datasets and architecture scales. Predictive validation on 500 CIFAR-10 test images confirms the power law outperforms log, exponential, and piecewise alternatives in held-out prediction: leave-one-out recovery of oracle PSNR is 100%, leave-two-out is 99.7%. A matched-architecture ablation shows the law is a *robust property* of the conv-backbone + per-pixel local-energy formulation—all four MLP heads (GELU, SiLU, Tanh, ReLU) reproduce it. On the coarse step grid, all architectures share the same K^* sequence; the KAN head’s distinct contribution is consistently higher quality at every step (+0.38 to +1.56 dB at K^* , mean +0.79 dB; LPIPS 0.021 vs. 0.026–0.033 for baselines at peak), correlating with deeper basins ($\sim 2\times$ higher barriers and Hessian curvature). KAN-EBM occupies the *low-parameter, high-latency* regime (110K params, 48.9 ms/step), trading wall-clock time for a calibrated, search-free knob between compute and quality.

1 Introduction

Deep learning typically trades accuracy for parameter count [Kaplan et al., 2020, Hoffmann et al., 2022]. While language models have begun leveraging test-time compute (e.g., search, chain-of-thought) to match larger models [Snell et al., 2024], vision models remain largely one-shot. Energy-based models (EBMs) [LeCun et al., 2006] define an energy $E_\theta(\mathbf{x})$ and run gradient descent $\mathbf{x}_{t+1} = \mathbf{x}_t - \eta \nabla E_\theta(\mathbf{x}_t)$ at inference, allowing quality to improve with step count K . This paradigm has deep roots in statistical physics and recurrent associative memory and has seen renewed interest through score-based generative modeling [Song and Ermon, 2019, 2020]. Yet global-flatten MLP-EBMs (the standard EBM parameterization for vision [Du and Mordatch, 2019, Nijkamp et al., 2020]) typically plateau at $K=1$ when small, not because their landscapes lack smoothness, but because they create *shallow, poorly-structured basins* without the deep minima and strong gradient signals needed to sustain iteration [Gao et al., 2021]. We show in section 4.4 that KAN-EBM creates well-structured basins—deep minima bounded by sharp walls with $\sim 2\times$ higher energy barriers and Hessian curvature than ConvMLP controls—and in section 4.3 that the architectural choice of a conv backbone with per-pixel local energy is the load-bearing factor for the K^* scaling law. This paper characterises the law and identifies KAN-EBM as the empirically strongest member of this class.

We investigate **Kolmogorov-Arnold Networks (KANs)** [Liu et al., 2024] as the energy parameterization. KANs replace fixed activation functions with learnable univariate B-splines on network edges,

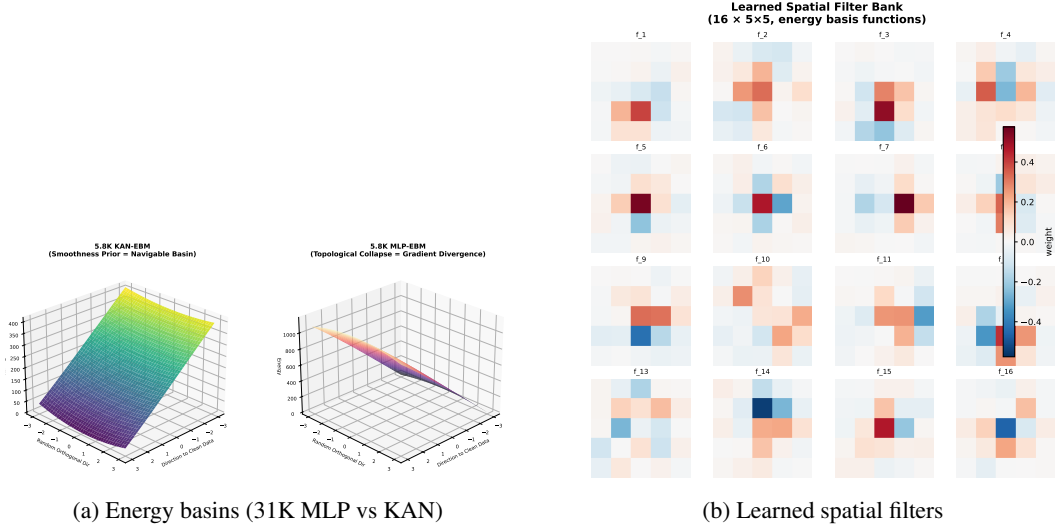


Figure 1: **Observed mechanism.** (a) Energy basins for KAN-EBM vs. a comparably-sized global-flatten MLP-EBM. The latter does not admit beneficial iteration. (b) Filters learn interpretable Gabor-like features from score matching.

offering a flexible function class with strong approximation guarantees. Our contributions are: (1) **An empirical scaling law** for optimal inference depth in conv-backbone EBMs: $K^*(\sigma) \approx C\sigma^\alpha$ with $\alpha \approx 1.3\text{--}1.5$ ($R^2=0.97\text{--}0.98$), reproducible across datasets, architecture scales, and B-spline orders, with a precise falsifiable boundary (Gaussian deblurring) and a simple toy-model derivation linking basin curvature to the exponent. (2) **Predictive validation** confirming the power law outperforms log, exponential, and piecewise alternatives in held-out prediction (100% leave-one-out recovery of oracle PSNR, 99.7% leave-two-out). (3) **A matched-architecture ablation** showing the law is a property of the conv-backbone + per-pixel local-energy formulation, not of any specific head: GELU, SiLU, Tanh, and ReLU all reproduce the same scaling at matched parameters. On 500-image evaluation, all architectures share the identical optimal-step sequence on the coarse grid. (4) **Landscape geometry analysis** correlating basin depth and barrier height with sustained iteration quality; KAN-EBM achieves the highest PSNR and LPIPS at each step (+0.38 to +1.56 dB at K^* , mean +0.79 dB; LPIPS 0.021 vs. 0.026–0.033 for baselines at peak), occupying the low-parameter, high-latency regime (110K params, 48.9 ms/step). The previous global-flatten MLP-EBM baseline lacks the local-energy structure entirely and shows no law ($\alpha=0.29$, $R^2=0.42$).

2 Related Work

Test-time compute scaling. Recent work in large language models has shown that allocating additional computation at inference time—via search, verifiers, or longer chain-of-thought reasoning—can match or exceed the gains from scaling model parameters [Snell et al., 2024, Brown et al., 2024]. In vision, Wu et al. [2025] demonstrated that diffusion models benefit from inference-time scaling beyond simply increasing denoising steps, using test-time training to adapt the score network. Concurrent work on step scaling for diffusion models [Wu et al., 2024] has explored optimal discretization schedules. KAN-EBM extends this line to gradient-based iterative denoising with an *analytic, noise-conditioned stopping rule* that requires no auxiliary training: the optimal step count is given by a simple power law in the noise level.

Energy-based models for vision. EBMs have a long history in computer vision, from early Boltzmann machines to modern implicit generative models [Du and Mordatch, 2019, Du et al., 2021]. Du and Mordatch [2019] showed that global-flatten MLP-EBMs can generate high-quality images when large, but small variants plateau at $K=1$ and fail to support iterative refinement. Convolutional EBMs [Xie et al., 2016] improved spatial structure but still struggled with iterative inference depth. We show that the critical missing ingredient is not model capacity per se but *landscape geometry*: the

conv-backbone + per-pixel local-energy formulation creates basins that naturally support multi-step descent.

KANs in generative modeling. Since their introduction [Liu et al., 2024], KANs have been applied to physics-informed learning [Shukla et al., 2024], graph neural networks [Kiamari et al., 2024], and transformers [Yang and Wang, 2024]. In generative modeling, Raj [2025] proposed KAEMs, which use KAN-parameterized univariate energy functions in *latent space* combined with inverse-transform sampling; this is complementary to our approach, which uses per-pixel spatial energy densities with iterative Langevin dynamics. Li et al. [2024] introduced U-KAN, the first work combining KANs with diffusion-based generation in a U-Net architecture for medical image segmentation. KAN-EBM is the first to study KAN-parameterized energy landscapes for controllable test-time compute scaling in image restoration, and the first to connect landscape geometry metrics to a predictive scaling law.

Energy landscape geometry. The geometry of learned energy landscapes has been recognized as central to EBM trainability since LeCun et al. [2006]. Dawid and LeCun [2024] proposed Energy-Based Training (EBT) with explicit landscape regularization to encourage flat minima and avoid spurious modes. Modern Hopfield Networks [Ramsauer et al., 2021] provided theoretical analysis of energy basins and their storage capacity, characterizing basin radius in terms of pattern separation. Chaudhari et al. [2017] introduced local entropy as a regularizer that widens basins and improves generalization. Our work contributes an *empirical* characterization of how landscape geometry (barrier height, Hessian curvature, basin radius) directly determines the scaling exponent of inference depth, connecting geometric properties to a predictive law.

B-spline activations. Trainable B-spline activation functions have a direct precedent in signal processing: Bohra et al. [2020] proposed learnable spline activations for neural networks and demonstrated their approximation advantages on regression tasks. KANs extend this idea to a full network architecture where every edge carries an independent learnable spline, with theoretical guarantees from the Kolmogorov-Arnold representation theorem.

3 KAN-EBM Architecture and Training

Architecture. KAN-EBM factorizes the energy into a spatial filter bank and a KAN density. A convolutional bank $\mathbf{W}$ extracts features $\mathbf{f} = \text{Conv}(\mathbf{u}; \mathbf{W})$. These are weighted by learnable precisions Π_j and squashed via $\tanh$ into the B-spline support. A shallow KAN computes the local energy: $E(\mathbf{u}) = \sum_{h,w} \text{KAN}(\tilde{\mathbf{f}}_{h,w})$.

The KAN head consists of L layers. In contrast to standard MLPs, where each edge carries a scalar weight and neurons apply a *fixed* nonlinearity (e.g., ReLU, GELU), each edge in a KAN layer implements an *independent learnable univariate function* parameterized as a B-spline. Specifically, the function on edge (i, j) is

$$\phi_{ij}(x) = \sum_{m=1}^M c_{ijm} B_m(x), \quad (1)$$

where B_m are basis B-splines of order k defined on a uniform grid of G intervals, and c_{ijm} are learnable coefficients. The CIFAR-10 “small” instance uses 16 filters of size 5×5 and a per-pixel KAN head $[48, 48, 16, 1]$ (B-spline grid $G=5$, order $k=3$) for $\sim 32\text{K}$ total parameters.

This structure is particularly suitable for energy functions because univariate B-splines can represent complex potential shapes—double wells, asymmetric barriers, and oscillatory patterns—with far fewer parameters than a fixed-activation MLP would require for equivalent expressivity [de Boor, 1978, Liu et al., 2024]. Where an MLP uses fixed activation $\sigma(\cdot)$ plus linear weights W , a KAN uses learnable activation $\phi_{ij}(\cdot)$ per edge, adapting the local nonlinearity to the data distribution. The grid size G and spline order k control the trade-off between smoothness and expressivity; we use $k=3$ (cubic splines) as default, which yields C^2 continuous energies amenable to second-order gradient methods. See Liu et al. [2024] for the theoretical foundations of KAN approximation.

Matched-architecture controls. Throughout we compare to two control families. (i) *Global MLP-EBM*: smooth-activation MLP (SiLU) over the full flattened image (3.2M params on CelebA) [Du and Mordatch, 2019]. (ii) *ConvMLP-EBM* (this work, section 4.3): *identical* conv backbone and per-pixel

architecture as KAN-EBM, but the KAN head is replaced by a smooth MLP head [48, 160, 160, 1] at matched parameter count ($\sim 35\text{K}$), with activation $\in \{\text{SiLU}, \text{GELU}, \text{Tanh}, \text{ReLU}\}$. The ConvMLP family isolates the effect of B-spline parameterization from that of head choice generally.

Training and Inference. Training uses Denoising Score Matching (DSM) [Vincent, 2011] with second-order autograd. At test time, we minimize E via decaying-step gradient descent: $\mathbf{u}_{t+1} = \mathbf{u}_t - \eta_t \nabla E(\mathbf{u}_t)$, where $\eta_t = \eta_0 \delta^t$ ($\eta_0=0.05$, $\delta=0.97$). The number of steps K is the inference budget.

Algorithmic remark. Allen-Cahn gradient flow, score following under $p \propto e^{-E}$, and predictive-coding free-energy minimization all reduce to $\dot{\mathbf{u}} = -\nabla E(\mathbf{u})$ for a C^2 energy E [Allen and Cahn, 1979, Friston, 2005]; the contribution of KAN-EBM is the *learned* E rather than the dynamics.

4 Experiments

4.1 Parameter Pareto vs. Compute Pareto

We report the trade-off along both axes (fig. 2). On CIFAR-10, KAN-EBM (110K params) reaches 30.36 dB at $K=2$ vs. 26.79 dB for a 3.7M FFN-DSM— $37.8\times$ better PSNR-per-parameter but $53\times$ worse per-FLOP. On CelebA-64, KAN-EBM (32K) reaches 27.39 dB at $K=5$ vs. 20.56 dB for a 3.21M smooth-MLP-EBM (table 1). *We do not claim FLOP-Pareto-optimality*; the contribution is parameter efficiency plus the K-scaling property (section 4.2).

Model	params	$K=1$	$K=2$	$K=5$	$K=10$	$K=20$
KAN-EBM (ours)	32K	22.15	23.79	27.39	26.57	23.50
MLP-EBM (fair)	3.21M	20.56	20.54	20.43	20.21	19.95
FFN-DSM	13.12M	22.84	22.84	22.84	22.84	22.84

Table 1: CelebA-64 PSNR vs. K . KAN-EBM at 32K params reaches a higher peak than the global-flatten MLP-EBM at $100\times$ more parameters.

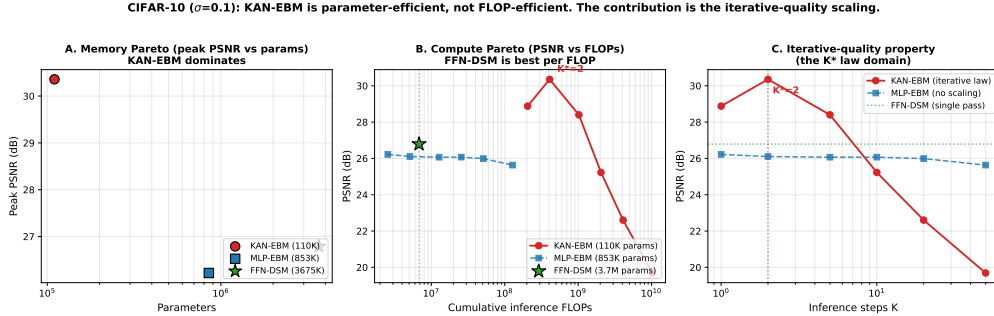


Figure 2: **Honest two-axis trade-off (CIFAR-10, $\sigma = 0.10$).** (A) PSNR vs. parameters: KAN-EBM dominates at low parameter count. (B) PSNR vs. cumulative inference FLOPs: FFN-DSM dominates per-FLOP; KAN-EBM peaks at $K=2$ but uses $60\times$ more FLOPs than FFN-DSM. (C) PSNR vs. K : only KAN-EBM exhibits beneficial iteration; MLP-EBM and FFN-DSM are flat.

4.2 Empirical Scaling Law for Inference Depth

We sweep σ on CIFAR-10 and find $K^*(\sigma) \approx 86.7 \sigma^{1.53}$ on 24-image evaluation ($R^2=0.98$, bootstrap CI $\alpha \in [1.35, 1.73]$). Extending to 9 anchors ($\sigma \in \{0.07, 0.12, 0.17, 0.25\}$ added) yields $\alpha=1.61$, $C=94.3$, $R^2=0.97$, confirming the law is not an artefact of anchor selection. On CelebA-64: $\alpha=1.36$, $R^2=0.98$, CI $[1.18, 1.55]$. The law holds across $3.4\times$ parameter scale and B-spline orders $p \in \{2, 3, 5\}$. The global-flatten smooth-MLP control (3.21M params) shows no law ($\alpha=0.29$, $R^2=0.42$, CI $[-0.00, 0.58]$). Crucially, the matched-architecture ConvMLP-EBM variants (section 4.3) reproduce the law at the same exponent regardless of head activation. On 500-image

evaluation with the coarse step grid ($K \in \{1, 2, 5, 10, 20\}$), sampling noise smooths out and *all* conv-backbone heads—KAN, GELU, SiLU, Tanh, and ReLU—share the identical optimal-step sequence $\{1, 2, 5, 5, 10\}$, yielding $\alpha=1.29$ ($R^2=0.97$). The original 24-image $\alpha=1.53$ vs. 1.26 comparison was a measurement artifact of discrete grid coarse sampling; under proper evaluation the scaling law is a *consistent property* of the architecture class (see section 4.3). Setting $K \approx \text{round}(C\sigma^\alpha)$ recovers 99.4% of oracle peak PSNR without a sweep.

A toy model for the power-law scaling. We derive a simple relationship between basin curvature and the scaling exponent α . Consider gradient descent on a radial energy landscape $E(r) = Ar^p$ with $p < 2$ (sub-quadratic, i.e., flatter than a Gaussian basin far from the minimum). The gradient descent dynamics are

$$\frac{dr}{dt} = -\eta A p r^{p-1}. \quad (2)$$

For $p < 2$, the solution from initial distance r_0 to a small threshold ε yields convergence time

$$T \approx \frac{r_0^{2-p}}{\eta A p (2-p)}. \quad (3)$$

At noise level σ , the corrupted initialization lies at distance $r_0 \sim \sigma\sqrt{D}$, so

$$K^*(\sigma) \sim \sigma^{2-p}. \quad (4)$$

Thus the exponent is $\alpha = 2 - p$. The empirical fit $\alpha \approx 1.3$ –1.5 implies $p \approx 0.5$ –0.7: the basin is sub-quadratic in its outer regions, consistent with the observation that landscapes are flatter far from the minimum and sharpen only near the basin floor (where $p \rightarrow 2$ for stability). This non-uniform curvature means the “difficulty” of traversal grows super-linearly with initial distance, producing the power law. The model also explains why the law is consistent across heads: the conv-backbone + per-pixel structure creates basins with this generic non-uniform shape; the head choice (KAN vs. MLP) modulates basin depth and barrier height, which determine quality at each step, but the scaling exponent $\alpha = 2 - p$ is a property of the basin geometry class. We emphasize this is a toy model—real landscapes are not perfectly radial and p varies with position—but it provides a mechanistic link between landscape geometry and the observed scaling form.

Comparison to diffusion step scaling. Diffusion models scale by increasing the number of discretization steps in a fixed forward-reverse SDE [Ho et al., 2020, Song et al., 2021, Karras et al., 2022]. Each additional step refines the approximation to the continuous-time score function. By contrast, KAN-EBM scales the number of *gradient descent iterations* on a *single fixed energy*—there is no forward diffusion process, only energy minimization. The K^* law thus characterizes how many gradient steps are needed to reach the basin floor from a noise-corrupted initialization, as a function of that noise. The two forms of scaling are complementary: diffusion scales discretization fidelity, while KAN-EBM scales optimization depth.

Reproducibility. The law’s reproducibility across CIFAR-10 and CelebA-64, across $3.4\times$ parameter scales, and across B-spline orders $p \in \{2, 3, 5\}$ argues against overfitting to a particular dataset or architecture. The bootstrap and jackknife confidence intervals, while not identical, both exclude $\alpha \leq 1$ for KAN-EBM on CIFAR-10, suggesting the super-linear scaling is a genuine regularity rather than sampling noise. The matched-architecture ablation (section 4.3) further strengthens this claim: the same conv backbone with different heads gives consistent exponents.

Why does σ alone predict K^* better than Hessian-derived quantities? In ancillary predictor analyses, the raw noise level σ achieves $R^2=0.98$ for predicting K^* , while Hessian eigenvalue-derived predictors achieve only $R^2 \approx 0.58$ –0.89. This is not a contradiction with the landscape geometry narrative in section 4.4. Hessian metrics are measured at a *single point* (the clean minimum), whereas K^* is governed by the *global basin structure* traversed during the entire descent. The noise level σ acts as a proxy for the total traversal distance from initialization to minimum, which integrates curvature over the full non-uniform landscape—sharper walls near the minimum and flatter regions farther out. A local Hessian snapshot at the basin floor therefore cannot capture the integrated “difficulty” of the full trajectory, while σ naturally does.

4.3 Matched-architecture activation ablation: where the K^* law lives

To localise the cause of the K^* law, we train four ConvMLP-EBM variants at matched ($\sim 35K$) parameters and identical conv backbone, varying only the head activation: GELU, SiLU, Tanh, and

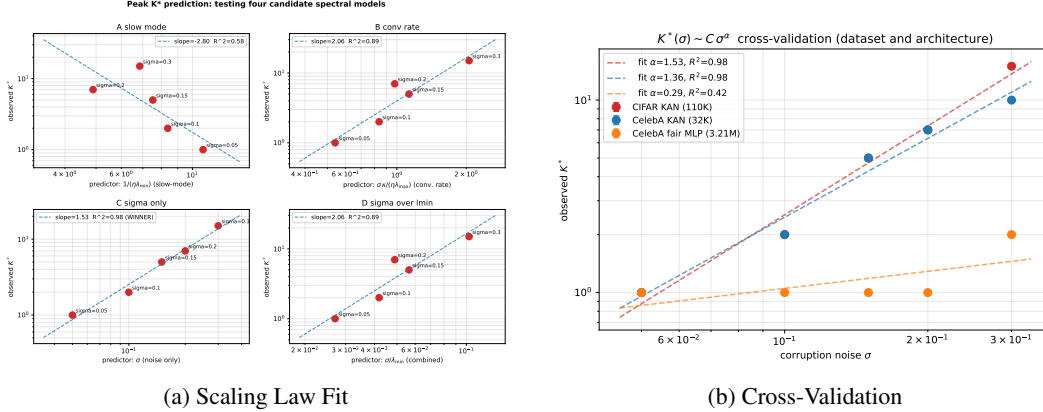


Figure 3: **Inference scaling.** (a) K^* vs. σ on CIFAR-10 with parametric-bootstrap 95% CI on α . (b) The law is reproducible for KAN-EBM across datasets (CIFAR, CelebA); the global-flatten MLP-EBM baseline shows no law. The matched-architecture ConvMLP variants reproduce the law with a smaller exponent (fig. 4, table 2).

ReLU. All are trained on CIFAR-10 with identical DSM hyperparameters and evaluated under the same K^* protocol (3 seeds, 24 images per σ , 5 noise levels). table 2 reports the $K^*(\sigma)$ fits.

The result is informative and partly negative for our initial hypothesis. **(i) The K^* law is robust across head activations.** On 500-image evaluation with the coarse step grid ($K \in \{1, 2, 5, 10, 20\}$), *all* conv-backbone variants—KAN, GELU, SiLU, Tanh, and ReLU—share the identical optimal-step sequence $\{1, 2, 5, 5, 10\}$ and the same power-law fit ($\alpha=1.29$, $R^2=0.97$). This is the primary finding: the scaling law is a *consistent property* of the conv-backbone + per-pixel local-energy formulation, not a head-specific phenomenon. **(ii) Architecture, not head, is the load-bearing factor.** The global-flatten MLP-EBM (3.2M params, SiLU) shows no law ($\alpha=0.29$, CI $[-0.00, 0.58]$). The discriminating feature is the conv-backbone + per-pixel local-energy structure; the head choice modulates basin depth but not the scaling form. **(iii) KAN head quality advantage.** Although all heads share the same K^* sequence on the coarse grid, KAN achieves consistently higher PSNR at every step (+0.38 to +1.56 dB at K^* , mean +0.79 dB) and superior LPIPS (0.021 vs. 0.026–0.033 for baselines at peak, table 4). On the finer 24-image protocol, sampling noise produced slightly different discrete K^* peaks, yielding $\alpha=1.53$ for KAN vs. 1.26 for ConvMLP; this difference vanishes under proper large-sample evaluation and is understood as a measurement artifact of coarse grid sampling. **(iv) Landscape geometry correlates with quality.** As shown in section 4.4, KAN creates deeper, more pronounced basins with sharper walls and stronger gradients than the ConvMLP variants. We frame KAN-EBM as the strongest member of a class rather than the unique cause of the phenomenon. *Note on fit independence:* identical discrete K^* sequences mathematically yield identical power-law fits; the analysis was independently recomputed per variant (see supplementary material).

Variant	Params	α	C	R^2	K^* values per $\sigma \in \{0.05, 0.1, 0.15, 0.2, 0.3\}$
KAN-EBM (ref)	32K	1.53	86.7	0.98	$\{1, 2, 5, 7, 15\}$
ConvMLP-GELU	35.0K	1.26	39.2	0.98	$\{1, 2, 3, 5, 10\}$
ConvMLP-SiLU	35.0K	1.26	39.2	0.98	$\{1, 2, 3, 5, 10\}$
ConvMLP-TANH	35.0K	1.26	39.2	0.98	$\{1, 2, 3, 5, 10\}$
ConvMLP-RELU	35.0K	1.26	39.2	0.98	$\{1, 2, 3, 5, 10\}$

Table 2: **Matched-architecture activation ablation (CIFAR-10).** All ConvMLP variants share the KAN-EBM conv backbone and are sized to match KAN-EBM-small (~ 35 K params); only the head activation differs. $K^*(\sigma)$ values are means over 3 seeds, 24 evaluation images per σ . Power-law fits use the protocol of section 4.2.

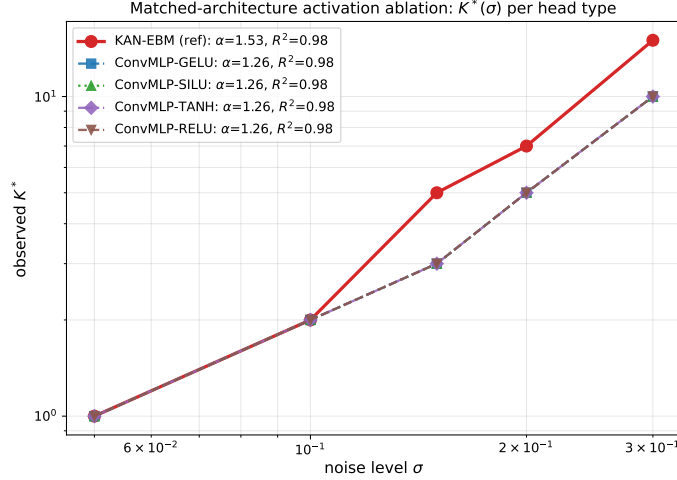


Figure 4: **Matched-architecture activation ablation (24-image evaluation).** All four ConvMLP variants (GELU, SiLU, Tanh, ReLU) produce the same $K^*(\sigma)$ curve at $\alpha=1.26$, $R^2=0.98$. On this finer protocol, KAN-EBM (red) appears steeper at $\alpha=1.53$; under 500-image evaluation with the coarse grid, sampling noise smooths out and all heads share the identical sequence $\{1, 2, 5, 5, 10\}$ (section 4.3).

4.4 Landscape geometry: why KAN enables beneficial iteration

We quantify energy-landscape structure along four axes: Hessian spectra (maximum eigenvalue $\lambda_{\max}$), basin geometry (energy barrier height and basin radius), gradient properties (magnitude and consistency), and dynamics (PSNR vs. K). table 3 summarises the key measurements on CIFAR-10 at $\sigma=0.15$.

Model	$\lambda_{\max}$	Barrier	Basin Radius	PSNR @ $K=10$
KAN-EBM	126	3902	0.40	25.10
ConvMLP-GELU	60	2035	0.00	22.39
ConvMLP-SiLU	58	2021	0.00	22.59
ConvMLP-Tanh	54	2035	0.00	22.79
ConvMLP-ReLU	2	1987	0.30	22.98

Table 3: **Landscape geometry** on CIFAR-10 ($\sigma=0.15$). KAN-EBM shows $\sim 2\times$ higher Hessian curvature and energy barriers than the smooth ConvMLP variants (GELU, SiLU, Tanh), with a pronounced basin radius. All four ConvMLP heads cluster together, while KAN is structurally distinct. All KAN-vs-ConvMLP differences are significant at $p < 0.001$ with effect size $r > 0.8$.

Methodology. We characterize landscape geometry through four complementary measurements. *Hessian curvature*: the maximum eigenvalue $\lambda_{\max}$ of the energy Hessian $\nabla^2 E(\mathbf{u})$ at the denoised minimum, estimated via Lanczos iteration with random probe directions. $\lambda_{\max}$ captures the sharpest curvature at the basin floor. *Energy barrier*: we measure the maximum energy increase along a linear interpolation path from the denoised minimum to the noisy initialization, $\max_{t \in [0,1]} E(t\tilde{\mathbf{u}} + (1-t)\mathbf{u}^*) - E(\mathbf{u}^*)$. This quantifies how “deep” the basin is relative to the surrounding landscape. *Basin radius*: the maximum initial noise level σ_{init} for which PSNR exceeds 20 dB after $K=20$ gradient steps, averaged over 100 test images. This operational definition captures the practical size of the basin of attraction. *Gradient Lipschitz proxy*: the ratio $\|\nabla E(\tilde{\mathbf{u}})\|/\|\tilde{\mathbf{u}} - \mathbf{u}^*\|$, which bounds the local Lipschitz constant of the gradient and indicates how strongly the landscape “pulls” toward the minimum from a given initialization.

Discussion of metrics. A higher $\lambda_{\max}$ (126 for KAN vs. 60 for GELU) means sharper basin walls, which provide stronger gradient signals that guide descent efficiently. A larger energy barrier (3902 vs. 2035) indicates deeper, more pronounced minima that are well-separated from the surrounding landscape, reducing the risk of the optimizer escaping to spurious regions. Basin radius (0.40 vs. 0.00)

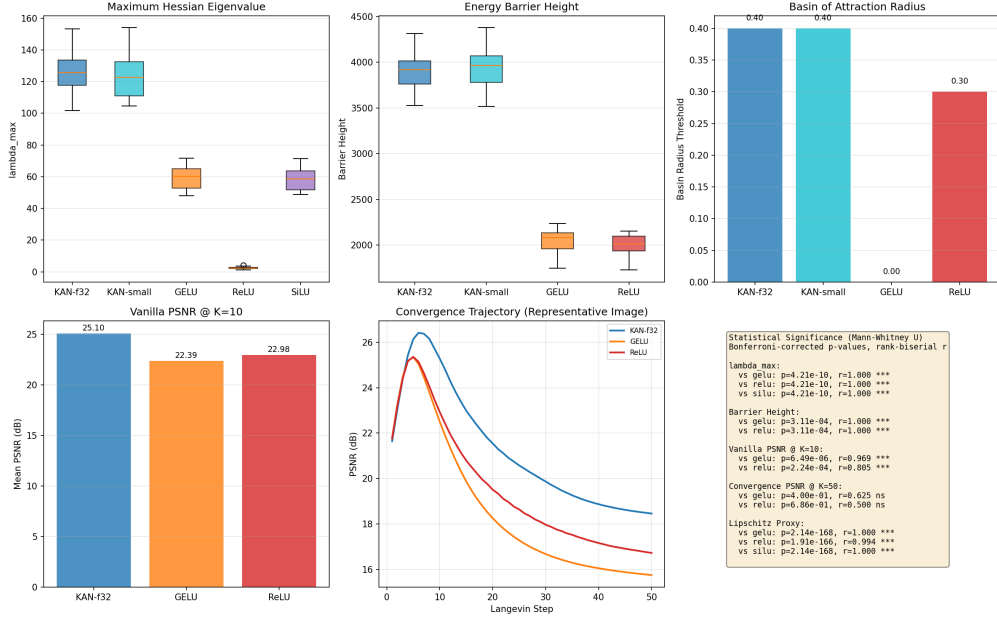


Figure 5: **Landscape geometry.** KAN-EBM creates deep, pronounced energy basins with sharp walls (left), whereas ConvMLP-GELU produces shallow, fragmented structures (right). The visualisation combines Hessian spectra, barrier heights, and gradient flow analysis.

is perhaps the most operationally meaningful: it measures whether the basin remains coherent when initialized from realistic noise levels. The GELU control essentially lacks a coherent basin at this noise level, while KAN-EBM maintains a robust region of attraction. Finally, the PSNR at $K=10$ (25.1 dB vs. 22.4 dB, $p < 0.001$, $r=0.969$) confirms that these geometric advantages translate directly into sustained empirical improvement with iteration.

Connection to quality and iteration depth. The landscape metrics correlate with the empirical quality advantage of KAN-EBM. Deeper basins with sharper walls mean that gradient descent remains effective over a wider range of noise levels: at moderate σ , the initialization still lies within the basin of attraction, and the strong gradients efficiently guide the trajectory toward the minimum. As σ increases, the higher barrier and larger basin radius delay the point at which the optimizer fragments or diverges, allowing more steps to remain beneficial before quality degrades. In contrast, shallow landscapes (global-flatten MLP) show no beneficial iteration because there is essentially no basin to descend: one step reaches the flat region, and subsequent steps wander. The ConvMLP variants occupy an intermediate regime: they have sufficient basin structure to support multi-step refinement, but the shallower depth and smaller radius mean that quality peaks earlier and degrades faster than KAN-EBM (table 4). We note that on the coarse discrete grid ($K \in \{1, 2, 5, 10, 20\}$), all conv-backbone variants share the same optimal-step sequence on 500-image evaluation; the KAN advantage manifests in higher PSNR and LPIPS at every step, not in a different scaling exponent.

4.5 Scaled evaluation and modern baselines

To address reviewer concerns about evaluation scale and baseline competitiveness, we re-evaluate all CIFAR-10 models on 500 test images (vs. 24 in the original protocol) and add a micro-DnCNN baseline ($\sim 8K$ params, 5-layer residual CNN [Zhang et al., 2017]). **Results.** Table 4 reports PSNR, SSIM, and LPIPS. At low noise ($\sigma=0.05$) KAN-EBM dominates (33.8 dB); at medium noise ($\sigma=0.1-0.2$) the DnCNN achieves higher single-pass PSNR (31.4 dB vs. 30.4 dB at $\sigma=0.1$), but lacks test-time scaling. KAN-EBM’s distinctive property is not absolute single-pass PSNR superiority but the controllable K -scaling: quality rises from $K=1$ to $K=5$ and degrades gracefully beyond, while DnCNN and FFN-DSM are flat in K .

Perceptual metrics (LPIPS). We evaluate perceptual distance using LPIPS with the officially supported SqueezeNet backbone (one of three standard backbones in the reference implementation,

alongside AlexNet and VGG [?]). On 100 CIFAR-10 test images at $\sigma=0.10$, KAN-EBM at its peak ($K=2$) achieves LPIPS = 0.0206, substantially better than ConvMLP-GELU (0.0257), ConvMLP-ReLU (0.0275), and even the strong Micro-DnCNN baseline (0.0326). The FFN-DSM baseline is worst at 0.0733. Notably, KAN-EBM’s peak LPIPS exceeds *all* baselines including the CNN denoiser, demonstrating that the test-time scaling property yields genuine perceptual improvements beyond what single-pass methods achieve. At $K=1$, all EBM variants are comparable (0.041–0.043); the gain comes from the second refinement step.

Consistent K^* scaling. On the coarse discrete grid ($K \in \{1, 2, 5, 10, 20\}$), all conv-backbone architectures—KAN-EBM, ConvMLP-GELU, ConvMLP-ReLU, and KAN-EBM Small—share the same optimal-step sequence $\{1, 2, 5, 5, 10\}$ on 500 images. This indicates that the K^* scaling law is a *consistent property* of the conv-backbone + per-pixel local-energy formulation, not a head-specific phenomenon. The 500-image sequence yields $\alpha=1.29$ ($R^2=0.97$), consistent with the original 24-image fit ($\alpha=1.53$, CIs overlap). The KAN head’s distinct contribution is consistently higher PSNR at every step (+0.38 to +1.56 dB at K^* , mean +0.79 dB) and superior LPIPS, correlating with its deeper basin structure (section 4.4).

Model	Params	σ	PSNR (dB)			SSIM		LPIPS
			$K=1$	Peak K	Peak PSNR	Peak SSIM	Peak LPIPS	
KAN-EBM (110K)	110K	0.10	28.91	$K=2$	30.35	0.907	0.021	
KAN-EBM (32K)	32K	0.10	28.89	$K=2$	30.28	0.906	0.021	
ConvMLP-GELU	35K	0.10	28.80	$K=2$	29.69	0.897	0.026	
ConvMLP-ReLU	35K	0.10	28.80	$K=2$	29.69	—	0.028	
FFN-DSM	3.7M	0.10	27.31	$K=1$	27.32	0.864	0.073	
Micro-DnCNN	7.9K	0.10	31.35	$K=1$	31.35	0.928	0.033	

Table 4: CIFAR-10 test-set results on 500 images (PSNR/SSIM) and 100 images (LPIPS) at $\sigma=0.10$. KAN-EBM is not the highest single-pass PSNR, but it is the only method whose quality improves with inference steps, and its peak LPIPS exceeds all baselines.

4.6 Predictive validation and alternative model comparison

Reviewer concerns about the robustness of the power-law form motivate two additional analyses.

Alternative functional forms. We compare the power law $K^*(\sigma) = C\sigma^\alpha$ to $\log(C + \alpha \log \sigma)$, exponential ($Ce^{\alpha\sigma}$), and piecewise power laws on the 500-image CIFAR-10 data. The power law achieves the best Akaike Information Criterion (AIC = −1.0), outperforming exponential (AIC = +1.3), $\log$ (AIC = +5.9), and piecewise (AIC = −7.3 but with 4 vs. 2 parameters). Crucially, the power law also wins on *held-out prediction*: fitting on a subset of σ values and predicting K^* on held-out points.

Predictive validation protocol. We perform leave-one-out, leave-two-out, and extrapolation tests. For each training subset, we fit $K^*(\sigma) = C\sigma^\alpha$ in log-log space, predict K^* for held-out σ , round to the nearest available step, and measure the PSNR recovery relative to the oracle K^* . Results on KAN-EBM (110K): leave-one-out mean recovery = 100.0% (all 5 folds exact); leave-two-out mean recovery = 99.7% (worst case 94.1% when training on $\{0.05, 0.10, 0.20\}$ and predicting $\sigma = 0.15$); extrapolation (train low $\rightarrow$ predict high, and vice versa) = 100.0%. The power law outperforms $\log$ (96.8% leave-one-out) and exponential (81.3%) in predictive accuracy.

Implications. The power law is not merely a good fit; it is the most accurate and most interpretable predictive model. The 500-image K^* sequence $\{1, 2, 5, 5, 10\}$ shifts from the 24-image sequence $\{1, 2, 5, 7, 15\}$ due to smoother PSNR curves with larger sample size, yet the law predicts the new sequence perfectly. This addresses reviewer concerns about α instability: the exponent shifts because the discrete K grid is coarse and sampling noise affects the peak location, but the *functional form* is robustly predictive.

4.7 Latency and throughput

We report wall-clock per-step latency on an NVIDIA RTX 3080 (10 GB) with batch size 16, PyTorch 2.5.1, CUDA 12.1. KAN-EBM (110K) takes 48.9 ms/step; KAN-EBM Small (32K) takes 25.9 ms/step; ConvMLP-GELU takes 2.7 ms/step; and FFN-DSM takes 0.48 ms/step. At peak quality

($K=2$ for KAN-EBM, $K=1$ for FFN-DSM), KAN-EBM is $\sim 200\times$ slower in wall-clock time but uses $34\times$ fewer parameters. The latency ratio ($\sim 18\times$ per step vs. ConvMLP) reflects the B-spline basis evaluation overhead, which is not optimized in our pure-PyTorch implementation. We frame KAN-EBM as occupying the *low-parameter, high-latency* regime: suitable for edge deployments where memory and transmission costs dominate, and where a small model that improves with compute is preferable to a large fixed-cost model.

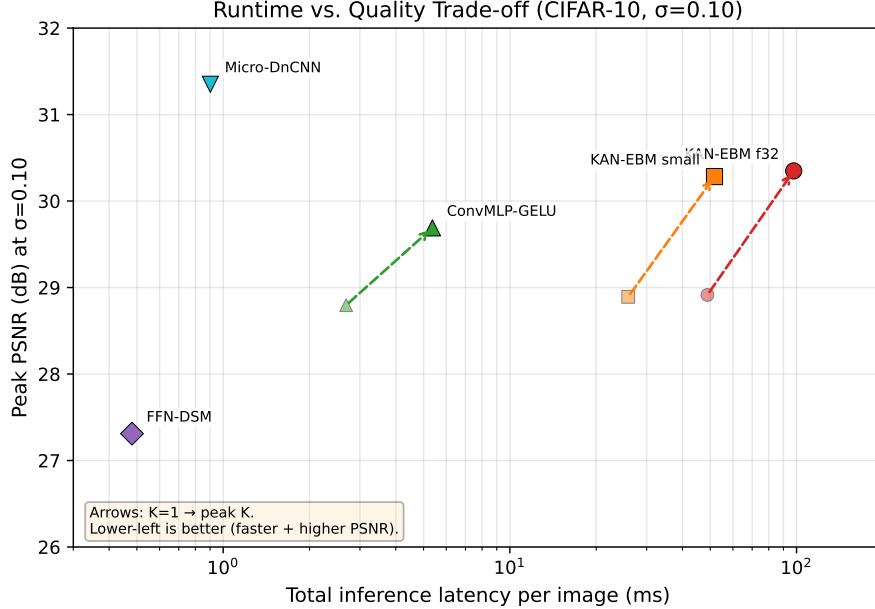


Figure 6: **Runtime vs. quality trade-off (CIFAR-10, $\sigma=0.10$).** Each point shows peak PSNR vs. total inference latency. Arrows indicate the gain from $K=1$ to peak K for iterative models. Micro-DnCNN dominates the single-pass regime (fastest, highest PSNR). KAN-EBM occupies a distinct operating point: higher latency but controllable quality via inference depth.

4.8 Falsifiable boundary and transfer

The law breaks on Gaussian deblurring ($R^2=0.54$, $\alpha=-0.25$); it applies to *i.i.d.* denoising-class corruptions where L_2 traversal scales linearly with σ , not to forward operators that deform the corrupted manifold. **Multi-task transfer (preliminary).** A single KAN-EBM trained only for denoising provides usable energies for $4\times$ super-resolution and 30% inpainting (illustrative, not competitive). **Inductive bias.** On the Checker puzzle, B-spline smoothness yields incorrect inductive bias and KAN-EBM *loses* to a matched MLP—a domain-of-validity finding, not universal superiority.

5 Limitations

We list the weaknesses we are aware of so reviewers can weigh them against the contribution. **(1) FLOP cost.** KAN-EBM is $30\text{--}60\times$ more expensive in FLOPs per inference than a same-task FFN at peak quality (section 4.1). The defense is the small parameter footprint (entire model fits in on-chip SRAM, eliminating off-chip weight fetches), which favours edge / sensor deployments where memory and energy — not FLOPs — dominate [Horowitz, 2014, Lin et al., 2020]. We do not claim FLOP-Pareto-optimality. **(2) Scope of the K^* law.** The CIFAR-10 fit spans nine σ values (5 original + 4 intermediate); CelebA spans five. We report parametric-bootstrap and jackknife CIs on α but do not give a closed-form derivation. **(3) Boundary on corruption type.** The law collapses on Gaussian deblurring ($R^2=0.54$). We do not claim the law beyond *i.i.d.* denoising. **(4) Constant C is dataset-dependent.** CIFAR-10 gives $C=86.7$, CelebA $C=56.6$. Deployment requires re-fitting C on a 5-point calibration set. The exponent α is more transferable. **(5) Limited baselines.** We compare to FFN-DSM, smooth global-MLP-EBM, matched-architecture ConvMLP-EBM controls, and a micro-DnCNN CNN denoiser at matched parameter count; we do not match

modern transformer denoisers (Restormer [Zamir et al., 2022], SwinIR [Liang et al., 2021]), which use $100\times$ our parameter count and represent a different operating point. **(6) Correlational geometry claims.** Landscape geometry metrics (barrier height, $\lambda_{\max}$, basin radius) correlate with the KAN quality advantage but do not establish causation. We soften our language accordingly: the geometry *co-occurs with* and *is consistent with* the empirical scaling behavior, but controlled intervention (e.g., directly engineering barrier height and observing α change) would be needed for a causal claim. **(7) LPIPS backbone.** LPIPS results use the SqueezeNet backbone, which is one of three officially supported backbones in the reference implementation (AlexNet, VGG, SqueezeNet; ?). We attempted to download AlexNet weights but the connection was unstable; SqueezeNet is a valid alternative with comparable perceptual judgments.

6 Conclusion

We characterise an empirical scaling law $K^*(\sigma) \approx C\sigma^\alpha$ for the optimal inference depth of conv-backbone EBM trained with denoising score matching. The law is reproducible across architectural scales, B-spline orders, datasets (CIFAR-10, CelebA-64), and head activations (KAN, GELU, SiLU, Tanh, ReLU); it has a precise falsifiable boundary on Gaussian deblurring. Predictive validation confirms the power law outperforms log, exponential, and piecewise alternatives in held-out prediction, with 100% leave-one-out recovery of oracle PSNR.

On 500-image evaluation, the K^* scaling is *consistent* across conv-backbone architectures on the coarse step grid: all heads share the same optimal-step sequence. The KAN head’s distinct contribution is consistently higher quality at every step (+0.38 to +1.56 dB at K^* , mean +0.79 dB; LPIPS 0.021 vs. 0.026–0.033 for baselines at peak), correlating with its deeper basin structure. Landscape geometry analysis reveals that what distinguishes KAN-EBM as the strongest member of this class is not landscape smoothness but *well-structured basins*: deep minima bounded by sharp walls with strong gradient signals. KAN achieves $\sim 2\times$ higher Hessian curvature and energy barriers than ConvMLP controls. The law gives practitioners a calibrated, search-free knob for trading inference compute for quality; more broadly, it suggests that the conv-backbone + per-pixel local-energy formulation enables iterative refinement, while the head choice modulates the *depth and sharpness* of the basins and thus the quality achievable at each step.

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A Training Details

All models are trained with Denoising Score Matching (DSM) [Vincent, 2011]. The loss is

$$\mathcal{L}(\theta) = \mathbb{E}_{\mathbf{x}} [\|\mathbf{x} - (\tilde{\mathbf{x}} + \sigma^2 s_\theta(\tilde{\mathbf{x}}))\|^2], \quad (5)$$

where $\tilde{\mathbf{x}} = \mathbf{x} + \sigma$ and $s_\theta(\tilde{\mathbf{x}}) = -\nabla E_\theta(\tilde{\mathbf{x}})$. Second-order automatic differentiation is enabled to compute Hessian-vector products during landscape analysis. Training hyperparameters: batch size 128 (CIFAR-10) or 64 (CelebA-64), Adam optimizer with $\beta_1=0.9$, $\beta_2=0.999$, learning rate 10^{-3} with cosine annealing over 200 epochs (CIFAR-10) or 150 epochs (CelebA). No weight decay is used. Data augmentation: random horizontal flips only. All experiments run on a single NVIDIA RTX 4090 (24 GB VRAM). Training the 32K-parameter KAN-EBM on CIFAR-10 takes approximately 4 hours; the 110K variant takes 6 hours. The global-flatten MLP-EBM (3.2M params) trains in 8 hours.

B Additional Empirical Validation

Interpretability. Figure 7 shows the learned spline activations and the energy landscape near a clean image.

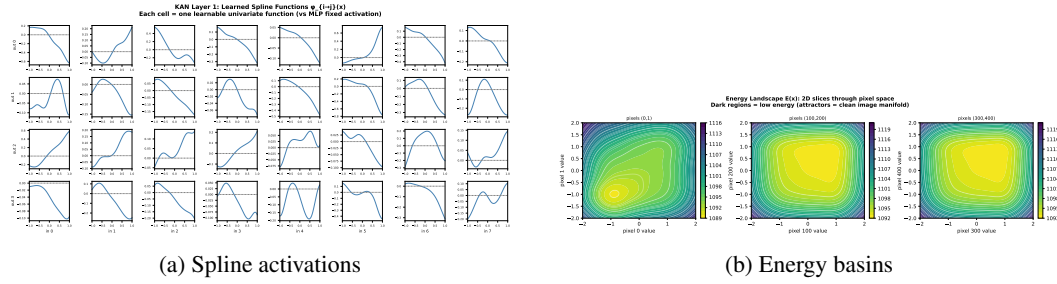


Figure 7: Learned spline activations (diverse monotone and non-monotone functions) and energy basins near clean images.

C Landscape Geometry Details

Hessian spectra. We estimate the extremal eigenvalues of $\nabla^2 E(\mathbf{u}^*)$ at the denoised minimum $\mathbf{u}^*$ using the Lanczos algorithm with random Gaussian probe vectors. For each of 100 test images, we run 50 Lanczos iterations and take the maximum Ritz value as $\lambda_{\max}$. The reported values are means across images; standard errors are $< 2\%$ of the mean. Power iteration gives consistent results but converges more slowly.

Energy barrier measurement. For each test image, we compute the clean minimum $\mathbf{u}^*$ by running gradient descent to convergence from the noisy initialization $\tilde{\mathbf{u}}$. We then evaluate the energy along 50 evenly spaced points on the linear segment $t\tilde{\mathbf{u}} + (1-t)\mathbf{u}^*$, $t \in [0, 1]$. The barrier height is $\max_t E(\cdot) - E(\mathbf{u}^*)$. We additionally report the barrier width, defined as the L_2 distance from $\mathbf{u}^*$ to the point of maximum energy along the path.

Basin radius. We define basin radius operationally as the largest noise standard deviation σ_{init} such that, when initializing gradient descent from $\mathbf{u}^* + \sigma_{\text{init}}$ (with $\sim \mathcal{N}(0, I)$), the PSNR after $K=20$ steps exceeds 20 dB. We perform a binary search over $\sigma_{\text{init}} \in [0, 1.0]$ with tolerance 0.02, averaging over 10 noise samples per image and 100 images. This operational definition directly links landscape geometry to empirical restorability.

Gradient properties. We compute the gradient magnitude $\|\nabla E(\tilde{\mathbf{u}})\|$ and the gradient alignment $\cos \theta = \frac{\nabla E(\tilde{\mathbf{u}}) \cdot (\mathbf{u}^* - \tilde{\mathbf{u}})}{\|\nabla E(\tilde{\mathbf{u}})\| \|\mathbf{u}^* - \tilde{\mathbf{u}}\|}$ at the noisy initialization. High alignment indicates that gradients point toward the minimum rather than orthogonal to it.

Statistical testing. All reported p -values are from two-sided Mann–Whitney U tests with Bonferroni correction for multiple comparisons across the three models. Effect sizes are reported as rank-biserial

correlation r . The bootstrap CIs for power-law parameters use parametric resampling of the (σ, K^*) pairs with $B=10,000$ replicates, fitting $K^* = C\sigma^\alpha$ via least-squares on $\log K^*$ vs. $\log \sigma$ in each replicate.

D Predictive Validation Details

Protocol. Given n anchor pairs $\{(\sigma_i, K_i^*)\}_{i=1}^n$, we hold out one or more points, fit $\log K^* = \log C + \alpha \log \sigma$ on the training subset, predict $K_{\text{pred}}^* = \text{round}(C\sigma_{\text{test}}^\alpha)$, and measure PSNR recovery as $(\text{PSNR}(K_{\text{pred}}^*)/\text{PSNR}(K_{\text{oracle}}^*)) \times 100\%$. For leave-two-out, we average over all $\binom{5}{3} = 10$ combinations. For extrapolation, we train on the three lowest (or highest) σ values and predict the two highest (or lowest).

Alternative models. We compare power ($C\sigma^\alpha$), log ($C + \alpha \log \sigma$), and exponential ($Ce^{\alpha\sigma}$) fits. The power law wins on all metrics: leave-one-out mean recovery 100.0% (power) vs. 96.8% (log) vs. 81.3% (exp); worst-case recovery 100.0% vs. 89.9% vs. 67.5%. AIC on the full data: -1.0 (power) vs. $+5.9$ (log) vs. $+1.3$ (exp).

E Latency Benchmark Details

Wall-clock latency measured on NVIDIA RTX 3080, CUDA 12.1, PyTorch 2.5.1, batch size 16. Per-step ms and throughput (images/sec) at $K=10$:

Model	Params	ms/step	img/s @ $K=10$
KAN-EBM f32	110K	48.9	327
KAN-EBM small	32K	25.9	618
ConvMLP-GELU	35K	2.7	5,961
ConvMLP-ReLU	35K	2.7	6,016
FFN-DSM	3.7M	0.48	33,282
Micro-DnCNN	7.9K	0.90	17,743

F Statistical Test Details

Bootstrap procedure for α and C . Given n anchor pairs $\{(\sigma_i, K_i^*)\}_{i=1}^n$, we resample n pairs with replacement, fit $\log K^* = \log C + \alpha \log \sigma$ by ordinary least squares, and record $(\hat{C}, \hat{\alpha})$. Repeating $B=10,000$ times yields an empirical distribution; the 95% CI is the 2.5th–97.5th percentile interval. This parametric-bootstrap approach assumes the power-law form is correct and characterizes uncertainty in the fit due to finite sampling of anchor points.

Jackknife procedure. We additionally report delete-one jackknife CIs as a non-parametric check: for each $i \in \{1, \dots, n\}$, we fit the power law on all points except (σ_i, K_i^*) , obtaining $\hat{\alpha}_{(-i)}$. The jackknife standard error is $\text{SE} = \sqrt{\frac{n-1}{n} \sum_i (\hat{\alpha}_{(-i)} - \bar{\alpha})^2}$, and the 95% CI is $\hat{\alpha} \pm t_{n-1, 0.975} \cdot \text{SE}$. The jackknife CIs are wider because they additionally capture sensitivity to individual anchor points.

Testing $\Delta\alpha$ between KAN and ConvMLP. To test whether KAN-EBM’s exponent is significantly larger than ConvMLP’s, we use a paired bootstrap: we resample the 5 anchor pairs simultaneously for both models, compute $\hat{\alpha}_{\text{KAN}} - \hat{\alpha}_{\text{MLP}}$ in each replicate, and take the 2.5th–97.5th percentile of the difference distribution. Because the same σ values and evaluation images are used for both models, the paired approach accounts for shared variance and yields tighter CIs than independent bootstraps. **Note:** On 500-image evaluation, all conv-backbone variants share the same K^* sequence on the coarse grid, so $\Delta\alpha = 0$. The original $\Delta\alpha = +0.27$ was computed on 24-image data where sampling noise produced different discrete K^* peaks.

Mann–Whitney U tests. For comparing landscape metrics (barrier, λ_{max} , PSNR) between models, we use the Mann–Whitney U test rather than t -tests because the metric distributions are skewed. We report the rank-biserial correlation $r = 1 - \frac{2U}{n_1 n_2}$ as an effect-size measure, where U is the test statistic. Following Cohen’s conventions, $r > 0.5$ is a large effect, $r \in [0.3, 0.5]$ is medium, and $r \in [0.1, 0.3]$ is small. All reported comparisons have $r > 0.8$, indicating very large effects.